ResiDual for Audio: Spectral Reweighting of Residual Streams in CLAP

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Abstract

Contrastive audio-language models such as CLAP achieve remarkable zero-shot classification performance, yet their internal representations remain largely unexplored. I investigate the structure of attention head outputs in HTS-AT, revealing a progressive emergence of semantically specialized representations across the network hierarchy, tightly linked to the intrinsic dimensionality of per-head activations. These insights motivate ResiDual*, a parameter-efficient method that reweights the principal components of each attention head without modifying any pretrained weight. Evaluated under 5-fold cross-validation on four benchmarks, ResiDual* substantially improves over the frozen CLAP baseline, with gains from $+2.4$ pp on ESC-50 to $+54.8$ pp on TinySOL.

1. Introduction

Contemporary audio–text encoders excel at capturing multimodal correspondences (**??**), yet the spectral structure expressed through attention heads remains comparatively under-analysed. Recent findings indicate that attention heads give rise to highly low-dimensional residual subspaces (**??**), suggesting latent geometric constraints that audio-domain Transformers do not explicitly exploit. This motivates an investigation into whether analogous structures emerge in HTS-AT (**?**) — a hierarchical Swin-Transformer variant that serves as the audio encoder in CLAP (**?**), mapping raw waveforms to a shared audio-text embedding space — and whether spectral selectivity can serve as an effective adaptation strategy for improving zero-shot classification.

In response, I introduce RESIDUAL*, a spectral reweighting framework extending the residual-subspace methodology of ResiDual (**?**) to the audio domain. The method (i) analyses attention-head dimensionality and semantic specialization across all 184 HTS-AT heads, revealing how representational complexity evolves across the network hierarchy and which stages are most amenable to spectral intervention; and (ii) yields measurable zero-shot classification improvements by learning, on a small labelled partition, a per-component rescaling of each attention head's principal directions — amplifying task-relevant spectral components and suppressing noise — without modifying any pretrained parameter. Full architectural details of CLAP and HTS-AT are provided in Appendix A; the complete head analysis pipeline and results in Appendix B; the ResiDual* implementation in Appendix C; and the full evaluation protocol and per-fold results in Appendix D. The appendices are intended to document every step of the research process — including design choices, intermediate experiments, and the reasoning that led to the final method.

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2. Related Work

Spectral Decomposition in Transformers. The ResiDual framework (**?**) formalizes residual streams through eigenspace decomposition, showing that attention heads operate within constrained, low-dimensional subspaces capturing specific semantic roles. This aligns with Voita et al. (**?**), who showed that only a subset of heads is critical for performance while others can be pruned with minimal impact.

Audio–Text Models and CLAP. CLAP (**?**) has emerged as a leading architecture for joint audio-text embeddings, building on CLIP-like methods (**?**) and attention-based audio models (**??**). Despite these advances, internal representation geometry within residual pathways remains underexplored: prior studies examined attention distributions (**???**) or audio embeddings (**?**), but not the spectral properties of residual streams. To the best of my knowledge, this is the first systematic spectral analysis and reweighting strategy applied to CLAP.

Spectral Reweighting and Representation Geometry. Pretrained models are known to produce anisotropic representations, degrading cosine-similarity-based comparisons (**?**). Post-hoc corrections such as All-but-the-Top (**?**)

and whitening (**?**) address this at the output embedding level. ResiDual (**?**) showed that reweighting principal components of attention head outputs — rather than output embeddings — yields more targeted spectral alignment. I extend this paradigm to the audio domain, applying it to the HTS-AT within CLAP.

3. Method

Baseline. The baseline is CLAP (**?**) used off-the-shelf in a fully frozen, zero-shot regime. Given an audio clip x, the ℓ_2-normalised audio embedding $\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$ is compared against the ℓ_2-normalised text embeddings $\{\mathbf{t}_c\}$ of class prompts `"this is the sound of {c}"`, and the predicted class is $\hat{y} = \arg\max_c \hat{\mathbf{a}}(x)^\top \mathbf{t}_c$. No adaptation, fine-tuning, or labelled data is used.[1]

Contribution: ResiDual*. ResiDual* adapts the frozen CLAP model by learning a per-component spectral reweighting of each attention head in HTS-AT, without modifying any pretrained parameter.

Head Analysis. To characterise the internal geometry of HTS-AT, I extract the raw pre-projection head outputs $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ for all 184 heads and spatially pool them into per-sample vectors:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N_w^\ell M} \sum_{i,j} \mathbf{H}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{d_h}. \tag{1}$$

I estimate intrinsic dimensionality via both linear (PCA, Effective Rank) and nonlinear (TwoNN, MLE) estimators, and measure semantic specialization through a cross-modal grounding framework that projects each head's principal axis into the CLAP embedding space and decomposes it via Orthogonal Matching Pursuit over a textual dictionary (see Appendix B).

Fitting, Optimisation, and Reweighting. For each target stage $\ell \in \mathcal{L}$, block b, and head h, I estimate a PCA basis $\mathbf{\Phi}_{\ell,b,h}$ and mean $\boldsymbol{\mu}_{\ell,b,h}$ from unlabelled clips via forward hooks, and initialise a learnable weight vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$. The vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are then optimised via cross-entropy minimisation on a small labelled partition. At inference time, each head output is transformed as:

$$\mathrm{RD}^*(\mathbf{H}_{\ell,b,h}) = \left(\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}\right) \mathbf{\Phi}_{\ell,b,h}\, \mathrm{diag}(\boldsymbol{\lambda}_{\ell,b,h})\, \mathbf{\Phi}_{\ell,b,h}^\top. \tag{2}$$

Full implementation details and hyperparameters are reported in Appendix C and Table 5.

[1] Since both vectors have unit norm, the dot product reduces to cosine similarity.

4. Results

Head Analysis. The dimensionality and semantic coherence analysis across all 184 heads (see Appendix B, Figures 3–4) reveals a clear structural transition: early-stage heads operate in near-unidimensional regimes ($\mathrm{EVR}_1 \approx 0.92$ in Stage 0), while Stages 2 and 3 develop high-dimensional, semantically coherent manifolds, with coherence Z-scores peaking at $Z = 1.57$ for heads such as `L3_B0_H7` (*human_body*) and `L2_B5_H15` (*nature*) (Figure 5, Table 4). This motivates restricting the reweighting target to $\mathcal{L} = \{2, 3\}$.

Zero-Shot Classification. Figure 1 reports mean 5-fold zero-shot accuracy across ESC-50, IRMAS, TinySOL, and VocalSound (full per-fold results in Table 6 and Table 7, Appendix D; hyperparameters in Table 5, Appendix C). ResiDual* consistently and substantially outperforms the frozen CLAP baseline on all benchmarks, with gains ranging from $+2.4$ pp on ESC-50 to $+54.8$ pp on TinySOL (see also Figure 6, Appendix D). The improvement is most pronounced on fine-grained timbral tasks (TinySOL, IRMAS), where the generic audio-language alignment of CLAP is insufficient, and the spectral reweighting successfully realigns internal representations toward task-relevant subspaces.

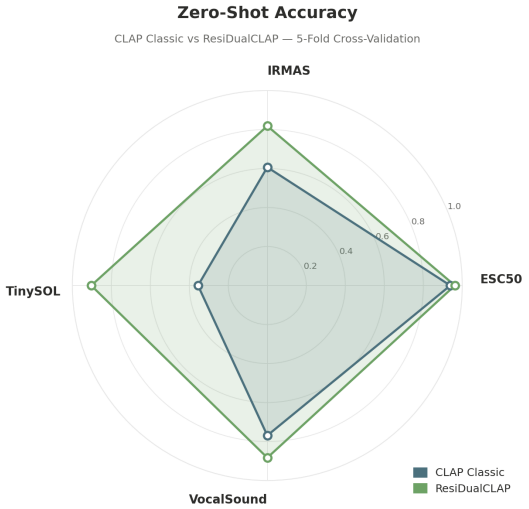


Figure 1. Mean 5-fold zero-shot accuracy across benchmarks. CLAP Classic (blue) vs. ResiDual* (green).

This pattern is consistent with the nature of the task: fine-grained datasets such as TinySOL and IRMAS are characterised by classes defined by highly distinctive timbral properties, localised in specific spectral regions. In such settings, a mechanism that selectively amplifies informative principal directions proves substantially more effective than the global, generic representations produced by the frozen CLAP encoder.

A. CLAP and HTS-AT: Architecture, Notation, and Residual Decomposition

This appendix is a self-contained reference for the two components at the core of our system. We begin by describing the CLAP dual-encoder model and its configuration (§A.1). We then present the HTS-AT audio backbone in detail, covering input preprocessing, hierarchical stage structure, and the internal mechanics of each attention block (§A.2). We then derive the per-head residual decomposition that underlies our analysis (§A.3). Finally, we provide implementation details for the ResiDual spectral reweighting (§C). All notation is introduced inline and collected in Table 3 for reference.

A.1. CLAP Overview

CLAP (Contrastive Language–Audio Pretraining) (**?**) is a dual-encoder model that aligns audio and text representations in a shared embedding space via contrastive learning. The audio encoder is HTS-AT (**?**) and the text encoder is GPT-2 (**?**) (embedding dimension 768), whose weights are frozen throughout training. Both encoders are independently projected to a joint space of dimension $d_{\text{proj}} = 1024$ via dedicated linear projection heads; similarity is measured by temperature-scaled cosine similarity with InfoNCE loss at temperature $\tau = 0.003$. Zero-shot audio classification is performed by comparing an audio embedding against the embeddings of textual class prompts.

The full configuration used in this work is reported in Table 1.

Table 1. CLAP configuration parameters.

Parameter	Value
Text encoder	GPT-2
Text encoder embedding dim	768
Audio encoder	HTS-AT
Audio encoder output dim (D_3)	768
Joint projection dim (d_{proj})	1024
Contrastive temperature τ	0.003
Sampling rate	44 100 Hz
Audio duration	7 s
Mel bands	64
FFT window	1024
Hop size	320
$f_{\min}$ / $f_{\max}$	50 Hz / 8 000 Hz

A.2. HTS-AT Architecture

HTS-AT (**?**) is a Swin-Transformer variant adapted for audio spectrograms. Its computation is organised into four hierarchical *stages*, each composed of several *blocks*, as illustrated in Figure 2. We describe the pipeline from raw audio to the final embedding in order: input preprocessing, patch embedding, stage structure, and the attention mechanism inside each block.

Input preprocessing and patch embedding. Each audio clip is converted to a 64-band log-mel spectrogram ($f_{\min} = 50$ Hz, $f_{\max} = 8{,}000$ Hz, STFT window 1024, hop 320) and normalised per mel-band. The resulting time–frequency matrix is rearranged into a square image $\mathbf{x}_{\text{img}} \in \mathbb{R}^{1\times 256\times 256}$ by folding the time axis into four contiguous segments of 256 frames stacked vertically over the 64 mel-frequency bands ($4\times 64 = 256$ rows, 256 columns). This image is then split into 4×4 non-overlapping patches via a strided `Conv2d` layer (kernel 4×4, stride 4), followed by `LayerNorm`, yielding the input token sequence

$$\mathbf{Z}^{(0,0)} = \mathbf{X}_{\text{in}} \in \mathbb{R}^{4096\times 96}, \tag{3}$$

where $4096 = 64 \times 64$ tokens each have embedding dimension $D_0 = 96$.

Stage structure and PatchMerging. The four stages operate at progressively coarser spatial resolutions, as summarised in Table 2. We index the residual stream as $\mathbf{Z}^{(\ell,b)}$, where $\ell \in \{0, 1, 2, 3\}$ is the stage index and $b \in \{1, \ldots, B_\ell\}$ is the block index within that stage.

After Stages 0, 1, and 2, a *PatchMerging* layer concatenates each 2×2 neighbourhood of spatially adjacent tokens into a single vector of dimension $4D_\ell$, then projects it to $2D_\ell$ via a bias-free linear layer. This halves the spatial side length S_ℓ and doubles the embedding dimension:

$$\mathbb{R}^{S_\ell^2\times D_\ell} \xrightarrow{\text{PatchMerging}} \mathbb{R}^{(S_\ell/2)^2\times 2D_\ell}. \tag{4}$$

Stage 3 has no PatchMerging. The resulting spatial side lengths are $S_\ell \in \{64, 32, 16, 8\}$ and the corresponding embedding dimensions are $D_\ell \in \{96, 192, 384, 768\}$ (see Table 2).

Within-stage residual structure. Inside each stage ℓ, both the attention and MLP sub-layers write additively to the residual stream. Following the pre-norm formulation (**?**), the stream after block b is:

$$\mathbf{Z}^{(\ell,b)} = \mathbf{Z}^{(\ell,0)} + \sum_{b'=1}^{b} \mathbf{A}_{\ell,b'} + \sum_{b'=1}^{b} \mathbf{M}_{\ell,b'}, \quad \in \mathbb{R}^{S_\ell^2\times D_\ell}, \tag{5}$$

where $\mathbf{Z}^{(\ell,0)}$ is the stage input, $\mathbf{A}_{\ell,b}$ is the W-MSA output, $\mathbf{M}_{\ell,b}$ is the MLP output, and S_ℓ is the spatial side length of the token grid at stage ℓ.

This decomposition holds strictly *within* a stage, where D_ℓ is constant. Across stage boundaries, PatchMerging

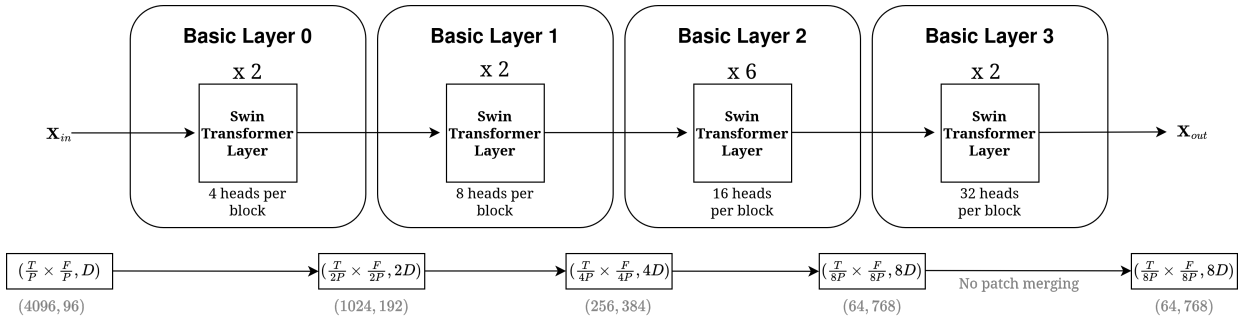


Figure 2. HTS-AT hierarchical architecture. The model consists of four stages of increasing embedding dimension: Stage 0 (2 blocks, 4 heads), Stage 1 (2 blocks, 8 heads), Stage 2 (6 blocks, 16 heads), and Stage 3 (2 blocks, 32 heads). PatchMerging is applied after Stages 0, 1, and 2, halving the spatial side length and doubling the feature dimension at each transition. Stage 3 has no PatchMerging. The input token sequence has spatial size $64 \times 64 = 4{,}096$ tokens with embedding dimension $D_0 = 96$; after three PatchMerging operations the final spatial size is $8 \times 8 = 64$ tokens with $D_3 = 768$.

Table 2. HTS-AT stage-level architectural parameters. The per-head dimension $d_h = D_\ell / H_\ell = 24$ is constant across all stages. S_ℓ is the spatial side length of the token grid at stage ℓ; $N_w^\ell = S_\ell^2/M$ is the number of attention windows, with $M = w^2 = 64$ tokens per window ($w = 8$).

Stage ℓ	**Blocks** B_ℓ	**Heads** H_ℓ	**Dim** D_ℓ	**Spatial** S_ℓ^2	**Windows** N_w^ℓ
0	2	4	96	64×64	64
1	2	8	192	32×32	16
2	6	16	384	16×16	4
3	2	32	768	8×8	1
Total heads H_{tot}	$2{\cdot}4 + 2{\cdot}8 + 6{\cdot}16 + 2{\cdot}32 = 8 + 16 + 96 + 64 = 184$				

changes both spatial resolution and embedding dimension ($S_\ell \to S_\ell/2$, $D_\ell \to 2D_\ell$), breaking any global additive structure across the full network, unlike in isotropic vision transformers (**?**).

Per-head output of the attention layer. The W-MSA output $\mathbf{A}_{\ell,b}$ is produced by concatenating the outputs of H_ℓ independent attention heads and projecting through a shared output matrix $W^O_{\ell,b}$. At block (ℓ, b), the raw output of head h is (see Appendix A.2 for derivation):

$$\mathbf{H}_{\ell,b,h} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{6}$$

where $N_w^\ell = S_\ell^2/M$ is the number of attention windows, $M = 64$ tokens per window, and $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M\times M}$ is the learned relative position bias for head h at block (ℓ, b) (shared across windows, independent per head and per block). All H_ℓ head outputs are concatenated and projected to give the W-MSA output (see Appendix A.3 for the full decomposition):

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \;\in\; \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}. \tag{7}$$

We analyse the raw head outputs $\mathbf{H}_{\ell,b,h}$ directly, *before* the output projection $W^O_{\ell,b}$. Each $\mathbf{H}_{\ell,b,h}$ lives in the per-head space $\mathbb{R}^{d_h}$, which is constant across all stages and blocks. This choice avoids entangling information across heads introduced by the shared projection, and enables uniform cross-stage comparisons of head geometry throughout the network. A complementary analysis of the projected contributions $\widehat{\mathbf{H}}_{\ell,b,h}$—which characterise how each head shapes the residual stream—is provided in Appendix **??**.

Block computation: W-MSA and MLP. Each block b at stage ℓ applies two sub-layers in sequence, both preceded by LayerNorm and connected via residual additions:

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b-1)} + \text{W-MSA}_{\ell,b}\Big(\mathrm{LN}\Big(\mathbf{Z}^{(\ell,b-1)}\Big)\Big), \tag{8}$$

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b)} + \mathrm{MLP}_{\ell,b}\Big(\mathrm{LN}\Big(\mathbf{Z}^{(\ell,b)}\Big)\Big). \tag{9}$$

Even-indexed blocks use standard Window Multi-head Self-Attention (W-MSA); odd-indexed blocks use Shifted-Window MSA (SW-MSA), which shifts the partition by $(w/2, w/2)$ tokens to enable cross-window interactions. At Stage 3, where $S_3 = w = 8$ so the grid coincides with a single window, the shift degenerates to zero and all blocks use W-MSA. The MLP sub-layer is a two-layer feed-forward network with hidden dimension $4D_\ell$ and GELU activation.

WindowAttention in detail. The W-MSA module at block (ℓ, b) first partitions the S_ℓ^2 tokens into $N_w^\ell = S_\ell^2/M$ non-overlapping windows of $M = 64$ tokens each, then applies multi-head attention independently within each window.

In the SW-MSA variant, a cyclic shift is applied before window partitioning.

Queries, keys, and values are computed via a single fused projection $W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$:

$$\mathrm{LN}(\mathbf{Z}^{(\ell,b-1)})\, W_{\ell,b}^{QKV} \xrightarrow{\mathrm{split}(D_\ell)} \mathbf{Q},\, \mathbf{K},\, \mathbf{V} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \tag{10}$$

Each tensor is then reshaped to isolate the H_ℓ heads, yielding per-head slices $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ with $d_h = D_\ell / H_\ell = 24$.

For each head $h \in \{1, \ldots, H_\ell\}$, attention is computed as:

$$\mathbf{H}_{\ell,b,h} = \mathrm{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{11}$$

where $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ is the learned relative position bias for head h of block (ℓ, b). This bias encodes the relative spatial offset between each pair of tokens *within* a window; it is shared across all N_w^ℓ windows (broadcast) and is independent for each head. Concretely, it is read from a parameter table of shape $((2w-1)^2, H_\ell) = (225, H_\ell)$ via a fixed index mapping, so each head h has its own column of learnable values.

All H_ℓ head outputs are then concatenated and passed through the block-specific output projection $W_{\ell,b}^O \in \mathbb{R}^{D_\ell \times D_\ell}$ with bias $\mathbf{b}_{\ell,b}^O \in \mathbb{R}^{D_\ell}$, giving the W-MSA output

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \;\in\; \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}, \tag{12}$$

which is added to the residual stream in Eq. (8).

A.3. Per-Head Residual Decomposition

The additive residual structure of Eq. (8) allows us to decompose the W-MSA contribution $\mathbf{A}_{\ell,b}$ exactly into independent per-head terms. Denoting by $W_{\ell,b,h}^O \in \mathbb{R}^{d_h \times D_\ell}$ the row slice of $W_{\ell,b}^O$ corresponding to head h (rows $[(h-1)d_h,\ hd_h)$), we distribute the output projection over heads:

$$\begin{aligned}\mathbf{A}_{\ell,b} &= \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \\ &= \sum_{h=1}^{H_\ell}\left(\mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \tfrac{\mathbf{b}_{\ell,b}^O}{H_\ell}\right) \;=\; \sum_{h=1}^{H_\ell} \widehat{\mathbf{H}}_{\ell,b,h},\end{aligned} \tag{13}$$

where we define the *per-head projected contribution*

$$\widehat{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \frac{\mathbf{b}_{\ell,b}^O}{H_\ell} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \tag{14}$$

This decomposition is exact and follows from the linearity of matrix multiplication: because $W_{\ell,b}^O$ acts on the concatenation of all head outputs, each head h effectively multiplies only its own row slice $W_{\ell,b,h}^O$, and the output bias can be distributed equally among heads without any approximation.

The decomposition holds within a single stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension, breaking any global additive structure; cross-stage comparisons must therefore account for this.

B. Head Analisys

B.1. Spatial Aggregation and Dataset Representations

Each raw head output $\mathbf{H}_{\ell,b,h}$ is a three-dimensional tensor indexed by window, token position, and feature dimension. To obtain a single fixed-size vector per audio sample, we mean-pool over all windows and token positions:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N_w^\ell\, M}\sum_{i=1}^{N_w^\ell}\sum_{j=1}^{M}\mathbf{H}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{d_h}. \tag{15}$$

This yields one vector $\mathbf{r}_{\ell,b,h}$ per audio sample, per head h, at every block (ℓ, b) of the network. Stacking these vectors across the n samples in the dataset gives the matrix

$$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}, \tag{16}$$

which is the primary object of our analysis. The total number of such matrices across the model is $H_{\mathrm{tot}} = 184$ (one per head per block; see Table 2).

Since $d_h = 24$ is constant across all stages, the ambient dimension of $\mathbf{R}_{\ell,b,h}$ is uniform throughout the network, enabling direct cross-stage comparisons of all dimensionality metrics without correction for a varying ceiling.

B.2. Dataset Representations

Per l'analisi della dimensionalità è stato scelto il dataset esc50 per la sua compattezza e per la sua ricchezza di classi. Infatti ...

B.3. Intrinsic Dimensionality Analysis

To characterize the effective complexity of the projected head representations $\{\mathbf{r}_{\ell,b,h}^{(i)}\}_{i=1}^n \subset \mathbb{R}^{1024}$, we employ a battery of linear and nonlinear dimensionality estimators.

B.3.1. Linear Estimators

PCA-based dimensionality. For each head, let $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ stack all aggregated representations. We compute

the covariance matrix $\mathbf{C}_{\ell,b,h} = \frac{1}{n-1}\mathbf{R}^\top\mathbf{R}$ and obtain ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$. Linear intrinsic dimensionality is:

$$d_{\mathrm{PCA}}(\alpha) = \arg\min_k \left\{ \frac{\sum_{i=1}^k \lambda_i}{\sum_i \lambda_i} \geq \alpha \right\}, \tag{17}$$

evaluated at $\alpha \in \{0.90, 0.95, 0.99\}$. We also report the *Explained Variance Ratio* of the first principal component, $\mathrm{EVR}_1 = \lambda_1 / \sum_i \lambda_i$.

Participation Ratio.

$$\mathrm{PR} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{18}$$

High PR signals uniform variance distribution; low PR signals dominance of few directions.

Effective Rank.

$$\mathrm{EffRank} = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{19}$$

which can be interpreted as the exponential of the spectral entropy. Unlike PCA thresholds, it provides a smooth, threshold-free notion of dimensionality, capturing how uniformly variance is distributed across components.

B.3.2. Nonlinear Estimators

TwoNN.

$$d_{\mathrm{TwoNN}} = \left(\frac{1}{n} \sum_{i=1}^n \log \frac{r_2^{(i)}}{r_1^{(i)}} \right)^{-1}, \tag{20}$$

where $r_1^{(i)}, r_2^{(i)}$ are Euclidean distances to the first and second nearest neighbours of $\mathbf{r}_{\ell,b,h}^{(i)}$ (**?**).

MLE.

$$\hat{d}_{\mathrm{MLE}}(\mathbf{r}) = \left(\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k(\mathbf{r})}{r_j(\mathbf{r})} \right)^{-1}, \tag{21}$$

averaged over all samples (**?**), with $k = 20$. This estimator is based on a local maximum likelihood principle, assuming uniform sampling in small neighbourhoods, and is sensitive to fine-scale manifold structure beyond global linear approximations.

Linear-Nonlinear (L/N) Ratio

$$\mathrm{Ratio}_B = \frac{\bar{d}_{\mathrm{PCA}_{99}}}{\bar{d}_{\mathrm{TwoNN}}}. \tag{22}$$

Values near 1 indicate near-linear manifolds; higher values signal nonlinear curvature beyond what PCA captures, and serve as a diagnostic for selecting layer targets for spectral reweighting.

B.4. Results: Dimensionality Evolution Across the Hierarchy

L'analisi della dimensionalità rivela una transizione strutturata nel modo in cui l'informazione acustica viene elaborata attraverso i blocchi di HTSAT. I risultati aggregati per blocco sono riportati in Figura 3, mentre il dettaglio per singola testa è illustrato in Figura 4.

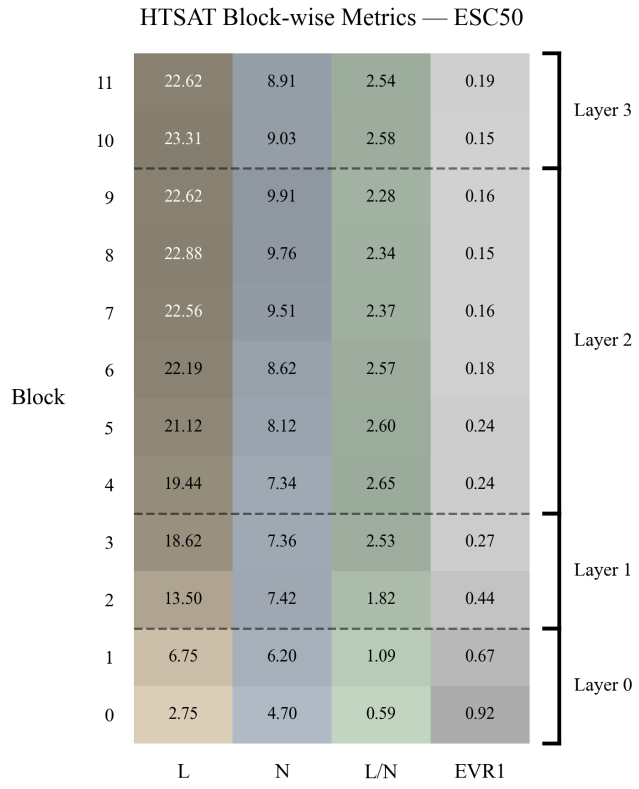


Figure 3. **Block-wise Metrics per il dataset ESC50.** La heatmap mostra l'andamento medio per ogni blocco (0-11) della dimensionalità lineare (L), nonlineare (N), del rapporto L/N e della dominanza della prima componente principale (EVR1). Le linee tratteggiate separano i quattro stage (Layer 0-3) dell'architettura.

B.4.1. Global Trends and Stage-wise Progression

Osservando la Figura 3, emerge chiaramente come la complessità delle rappresentazioni aumenti progressivamente risalendo la gerarchia del modello.

- **Layer 0 (Expansion Phase):** Nei primi blocchi, la dimensionalità intrinseca è minima ($L \approx 2.75$ nel

blocco 0). L'altissimo valore di EVR1 (0.92) indica che quasi tutta la varianza è concentrata in una singola direzione. Questo suggerisce che i primi stadi agiscono come estrattori di feature a bassa complessità, focalizzandosi su segnali dominanti del dominio tempo-frequenza.

- **Layer 1 & 2 (Complexity Growth):** Si osserva una crescita rapida di L e N. Il rapporto L/N raggiunge il suo picco nel Layer 2 (2.65), suggerendo che in questa fase la curvatura del manifold è massima. Le rappresentazioni non solo occupano più dimensioni, ma lo fanno in modo significativamente non lineare rispetto alle basi PCA.
- **Layer 3 (Saturation and Refinement):** Negli ultimi blocchi (10-11), la dimensionalità tende a stabilizzarsi o a subire una leggera contrazione (specialmente in N). La EVR1 crolla ai minimi storici (0.15), indicando una distribuzione della varianza estremamente uniforme tra le componenti disponibili ($d_h = 24$).

B.4.2. Head-specific Analysis and Manifold Structure

L'analisi a livello di singola testa (Figura 4) permette di apprezzare l'eterogeneità funzionale all'interno dello stesso blocco.

I grafici evidenziano alcuni fenomeni chiave:

- **Saturation of Linear Capacity:** Il panel (A) mostra che molte teste nei Layer 2 e 3 raggiungono la dimensionalità massima consentita dalla soglia $\alpha = 99\%$, saturando vicino al limite dell'ambient dimension ($d_h = 24$).
- **Effective Rank vs. PR:** Sia l'Effective Rank (E) che il Participation Ratio (D) confermano una transizione da un regime a "basso rango" (Layer 0) a un regime ad "alto rango" nei layer profondi. Tuttavia, la dispersione nel Layer 2 suggerisce che alcune teste mantengano una specializzazione più focalizzata (bassa dimensionalità) rispetto ad altre che esplorano lo spazio intero.
- **PC1 Dominance Decay:** Il panel (F) illustra la "decostruzione" della dominanza spettrale. Mentre le prime teste (Layer 0) hanno una PC1 che spiega quasi il 100% della varianza, nelle teste finali tale valore scende sotto il 20%, un segnale di rappresentazioni distribuite e potenzialmente più robuste.

Reflections on Representational Complexity. Questa evoluzione suggerisce che HTSAT operi una progressiva *unfolding* dei dati audio: da proiezioni iniziali quasi unidimensionali (probabilmente legate a feature energetiche o spettrali grezze) verso manifold complessi e multidimensionali nei layer intermedi, dove avviene la discriminazione semantica delle classi di ESC50. L'alto rapporto L/N nei Layer 2 e 3 identifica questi blocchi come i candidati ideali per interventi di regolarizzazione spettrale, essendo i punti di massima compressione non lineare dell'informazione.

B.5. Head Specialization Analysis

Per interpretare il contenuto semantico codificato dalle singole teste, introduciamo un framework di *cross-modal grounding* basato su un modello CLAP pre-addestrato. L'obiettivo è mappare l'asse di varianza principale di ogni testa in uno spazio di concetti testuali descrittivi.

B.5.1. Semantic Mapping via Principal Component Projection

Sia $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ la matrice delle rappresentazioni aggregate per una data testa. Per identificare la caratteristica acustica primaria catturata dalla testa, calcoliamo la prima componente principale $\mathbf{v}_1 \in \mathbb{R}^{d_h}$. Per ogni campione i, lo score sulla prima componente è definito come:

$$s_i = \mathbf{r}^{(i)}_{\ell,b,h} \cdot \mathbf{v}_1. \tag{23}$$

Definiamo quindi la **Direzione Semantica** $\sigma_{\ell,b,h}$ nello spazio latente CLAP come la combinazione lineare degli embedding audio CLAP $\{\mathbf{a}_i\}_{i=1}^n \subset \mathbb{R}^{D_{\text{CLAP}}}$, pesata dagli score normalizzati:

$$\sigma_{\ell,b,h} = \sum_{i=1}^{n} w_i \mathbf{a}_i, \quad \text{con } w_i = \frac{s_i - \bar{s}}{\|\mathbf{s} - \bar{s}\|_2}, \tag{24}$$

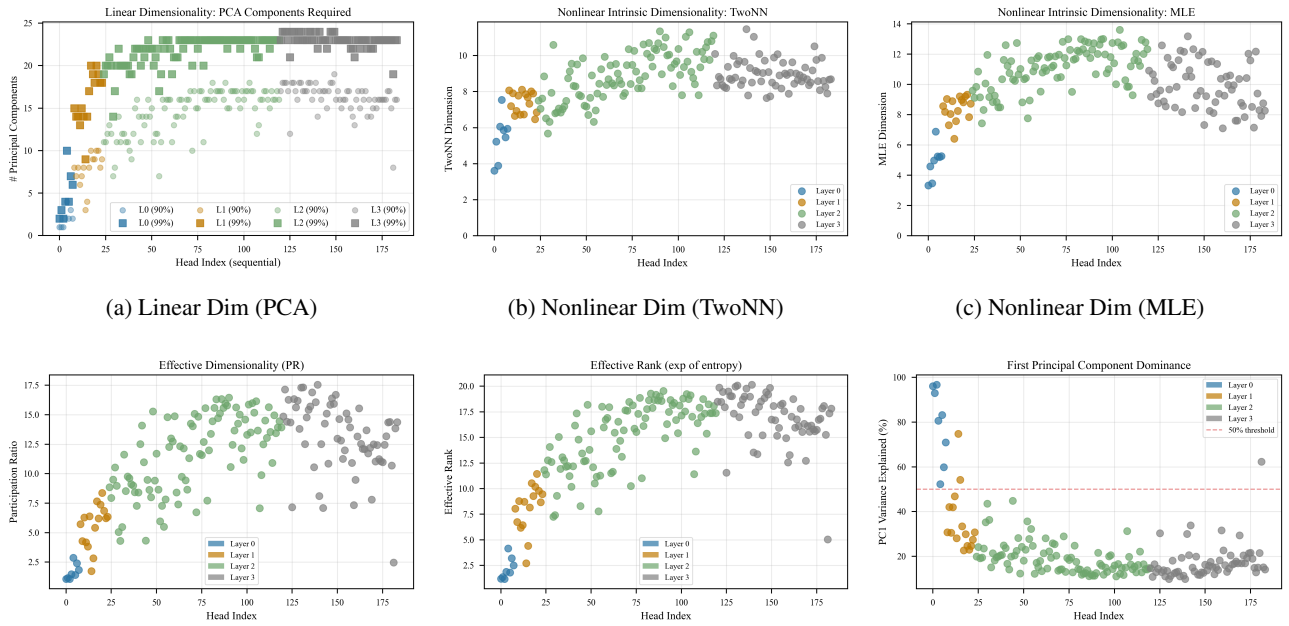
dove $\bar{s}$ è la media degli score. Questo vettore σ rappresenta l'asse di massima variazione della testa proiettato in uno spazio allineato al linguaggio naturale.

B.5.2. Sparse Coding via Orthogonal Matching Pursuit (OMP)

Per tradurre $\sigma_{\ell,b,h}$ in termini comprensibili, utilizziamo un dizionario testuale $\mathcal{D}$ composto da M descrizioni audio (es. "dog barking", "high pitched tone"). Siano $\{\mathbf{t}_j\}_{j=1}^M$ i corrispondenti embedding testuali CLAP. Poiché una testa potrebbe rispondere a una combinazione di concetti, decomponiamo σ mediante un algoritmo *Orthogonal Matching Pursuit* (OMP). L'OMP seleziona iterativamente i k atomi dal dizionario che meglio spiegano la direzione semantica, risolvendo:

$$\min_{\mathbf{x}} \|\sigma_{\ell,b,h} - \mathbf{T}\mathbf{x}\|_2 \quad \text{s.t. } \|\mathbf{x}\|_0 \leq k, \tag{25}$$

dove $\mathbf{T}$ è la matrice degli embedding del dizionario. Nel nostro esperimento fissiamo $k = 5$. Questo approccio "greedy" permette di distinguere tra teste altamente specializzate (che puntano a un singolo atomo) e teste che catturano concetti acustici più ampi.



(a) Linear Dim (PCA) (b) Nonlinear Dim (TwoNN) (c) Nonlinear Dim (MLE)

(d) Participation Ratio (e) Effective Rank (f) PC1 Dominance

Figure 4. **Analisi granulare delle 184 teste di HTSAT.** I panel mostrano la distribuzione delle metriche di dimensionalità per ogni testa, indicizzate sequenzialmente. (A-C) Confronto tra stime lineari e non lineari; (D-E) Metriche di diffusione spettrale; (F) Concentrazione della varianza sulla prima componente.

B.5.3. Metrics for Semantic Coherence

Per quantificare la qualità della specializzazione, introduciamo lo **Z-score di Coerenza**. Data la sottorete di k descrizioni selezionate $\mathcal{D}_k \subset \mathcal{D}$, definiamo la similarità intra-gruppo S_{intra} come la media dei prodotti scalari tra gli embedding selezionati. Lo Z-score è calcolato come:

$$Z = \frac{S_{\text{intra}} - \mu_{\mathcal{D}}}{\sigma_{\mathcal{D}}}, \tag{26}$$

dove $\mu_{\mathcal{D}}$ e $\sigma_{\mathcal{D}}$ sono rispettivamente la media e la deviazione standard delle similarità tra gli atomi selezionati e l'intero dizionario $\mathcal{D}$. Un valore di Z elevato indica che la testa è focalizzata su un dominio semantico ben definito e consistente (es. solo suoni naturali o solo parametri timbrici), mentre valori bassi indicano una specializzazione spuria o eccessivamente eterogenea.

B.6. Results: Semantic Landscapes and Head Specialization

L'applicazione del framework di *cross-modal grounding* rivela l'emergere di una specializzazione funzionale che segue la gerarchia strutturale del modello. L'analisi dello Z-score di coerenza semantica (Figura 5) e l'identificazione delle teste più specializzate (Tabella 4) permettono di capire come i concetti acustici astratti si cristallizzino in categorie semantiche.

B.6.1. Emergence of Semantic Coherence

L'andamento della coerenza semantica attraverso i layer (Figura 5) rivela una transizione di fase legata alla struttura gerarchica del modello:

- **Layer 0 e 1 (Semantic Void):** Nei primi quattro blocchi (0-3), la coerenza semantica è prevalentemente negativa o prossima allo zero. In questa fase, le teste estraggono feature a bassa dimensionalità ($L < 14$) e altissima dominanza spettrale ($\text{EVR}_1 \approx 0.92$). La bassa coerenza i ndica che queste teste rispondono a proprietà fisiche del segnale comuni a molteplici sorgenti, rendendo impossibile un mapping univoco su descrizioni testuali specifiche.

- **Layer 2 (Functional Diversification):** This stage marks the transition into a regime of increased nonlinearity. In particular, a clear increase in head specialization emerges toward the end of the layer, in concomitance with the growth of the intrinsic nonlinear dimensionality (as measured by TwoNN, Fig. 3) and the concurrent decrease in the leading eigenvalue share (EVR_1), which drops from 0.24 at the beginning of Layer 2 to 0.16 at its end (Fig. 3).

  This joint behaviour suggests that the expansion of the underlying manifold complexity is closely linked to the emergence of more semantically selective atten-

tion heads. Accordingly, highly specialized heads begin to appear, such as `L2_B4_H1` ($Z = 1.46$, absolute block 8), which capture fine-grained acoustic events associated with the *human_body* category.

- **Layer 3 (Semantic Convergence):** This layer initially continues the trend observed at the end of Layer 2, with a high density of more specialized heads in the early blocks of Layer 3. In this regime, heads remain well-aligned with CLAP semantic concepts, despite the fact that the leading eigenvalue share (EVR_1) reaches one of its lowest values (0.15), indicating that information is distributed across a larger number of latent components.

  However, this trend progressively reverses toward the final blocks (10–11), where a mild decrease in head specialization is observed. This reversal is consistent with a broader inversion in all structural metrics, including L, N, L/N, and EVR_1, which collectively shift from their previously increasing or decreasing trends (see Fig. 3).

  Overall, Layer 3 appears to exhibit an initial phase of strong semantic alignment, followed by a late-stage relaxation of specialization, suggesting a partial remixing of representations in the deepest blocks.

B.6.2. Identification of Specialized Heads

La Tabella 4 riporta una selezione delle teste che mostrano la massima coerenza semantica, evidenziando una predilezione per categorie con forti connotati timbrici o temporali.

The results reported in Table 4 confirm the emergence of a clear semantic organization of attention heads across layers. In particular, the highest coherence scores are consistently associated with heads that capture well-defined acoustic categories such as *human_body*, *nature*, and *urban sounds*, suggesting that specialization is not random but structurally aligned with meaningful audio concepts.

A notable trend is the predominance of highly coherent heads in Layers 2 and 3, supporting the observation that semantic specialization intensifies after the initial low-level feature extraction stage. Moreover, heads with higher Z-scores tend to exhibit lower PC1 variance, indicating that stronger semantic alignment correlates with more concentrated spectral structure.

Finally, the presence of moderately coherent but still structured heads in later blocks suggests that, while full specialization emerges primarily in intermediate layers, the deepest representations undergo a partial re-mixing of features, consistent with the non-monotonic behaviour observed in Fig. 5 and Fig. 3. Importantly, the observed inversion of the trend in the deepest blocks is also consistent with the findings reported in (**?**), indicating a shared structural behaviour across architectures and suggesting that this non-monotonic specialization pattern extends beyond the audio domain to image-domain residual stream models as well.

C. Reweighting Pipeline

C.1. ResiDual: Implementation Details

This section describes the full implementation of ResiDual (RD*), a learnable spectral adaptation integrated into a frozen CLAP model. The pipeline consists of two sequential phases: a *fitting pass* that estimates a fixed PCA basis for each attention head (§C.1.1), followed by an *optimisation phase* in which the per-head weight vectors are refined as free parameters via gradient descent to maximise zero-shot classification accuracy (§C.1.2). The forward-pass reweighting applied at inference time is described in §C.1.3.

Throughout this appendix we assume batch size $B = 1$ for notational clarity; all operations extend straightforwardly to $B > 1$ by treating the batch axis independently. We use the index conventions of Appendix A: $\ell \in \{0, 1, 2, 3\}$ is the stage index, $b \in \{1, \dots, B_\ell\}$ is the block index within stage ℓ, and $h \in \{1, \dots, H_\ell\}$ is the head index. The per-head feature dimension is $d_h = 24$ (constant across all stages), $N_w^\ell = S_\ell^2/M$ is the number of attention windows at stage ℓ, and $M = 64$ is the number of tokens per window. All symbols are collected in Table 3.

Spectral reweighting layers. For each target stage $\ell \in \mathcal{L}$, block b, and head h, RD*instantiates a `SpectralReweightingLayer` — a lightweight `nn.Module` that carries two non-trainable buffers, $\mathbf{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ and $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$, and a single learnable parameter vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$. All such modules are collected in a nested `nn.ModuleDict` indexed by (ℓ, b, h), constructed once during model initialisation in `_build_spectral_layers` from the stage dimensions and head counts of the underlying HTS-AT configuration; they persist in the model's `state_dict` across saves and loads. Each module exposes two methods: `fit_pca`, which populates $\mathbf{\Phi}_{\ell,b,h}$ and $\boldsymbol{\mu}_{\ell,b,h}$ from data and initialises $\boldsymbol{\lambda}_{\ell,b,h}$ (§C.1.1); and `forward`, which applies the RD*transform (§C.1.3).

Module injection via forward hooks. RD*injects its reweighting through permanent PyTorch forward hooks. For each target block (ℓ, b) with $\ell \in \mathcal{L}$, a dedicated `AttentionHook` object is registered once on the corresponding `WindowAttention` module at model construction via `register_forward_hook` and remains attached throughout the entire model lifetime — no regis-

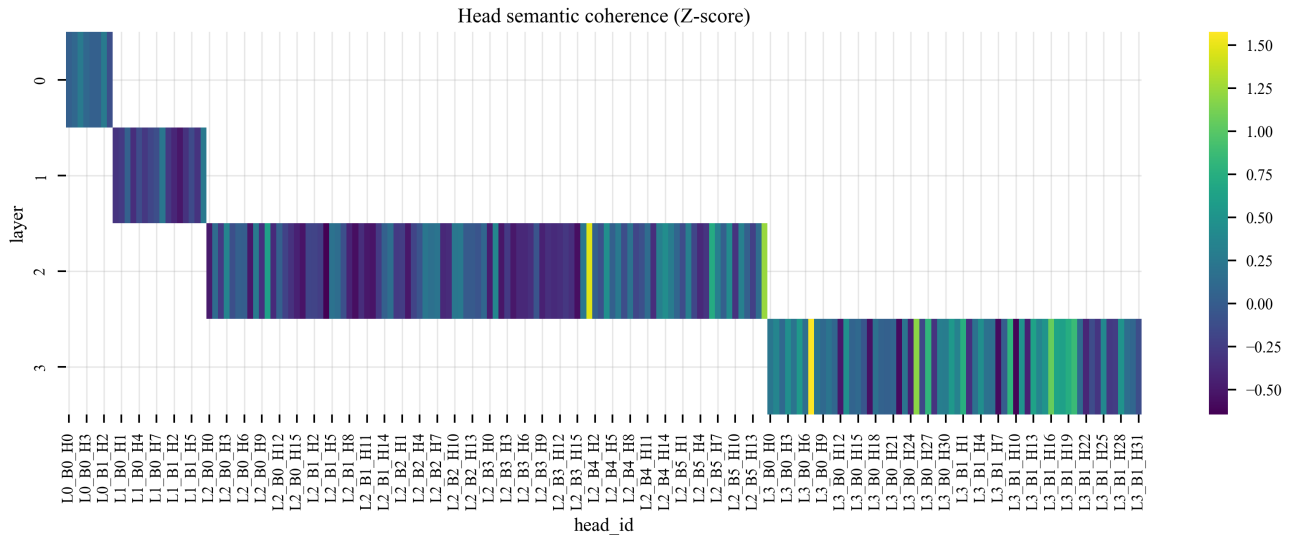


Figure 5. **Heatmap dello Z-score di coerenza semantica.** La mappa evidenzia come la capacità di allineamento semantico emerga drasticamente a partire dalla fine del Layer 2, in corrispondenza dell'aumento della dimensionalità non-lineare e della riduzione della dominanza della prima componente principale.

tration or removal occurs at inference time. The hook is stateless: it holds only a reference to the corresponding slice `spectral_layers`$[\ell][b]$ of the module dict, and operates in one of two modes controlled by a `collect_for_fitting` flag. In *collection mode*, it accumulates the raw per-head tensors $\mathbf{H}_{\ell,b,h}$ without modifying the forward-pass output, feeding the fitting pass of §C.1.1. In *reweighting mode*, it replaces $\mathbf{H}_{\ell,b,h}$ with the reweighted output $\mathbf{H}^*_{\ell,b,h}$ (Eq. (36)) and immediately re-applies the output projection $W^O_{\ell,b}$ and dropout `proj_drop` within the hook itself before returning the modified output; this is necessary because returning a modified tensor from the hook bypasses the projection that would otherwise follow in the original `WindowAttention` forward. No structural modification of the original model is performed: all pretrained parameters — the fused QKV projection $W^{QKV}_{\ell,b}$, the output projection $W^O_{\ell,b}$ with bias $\mathbf{b}^O_{\ell,b}$, and the relative position bias table $\mathbf{B}_{\ell,b,h}$ — remain exactly as loaded from the checkpoint.

C.1.1. Fitting Pass: PCA Basis Estimation

Data collection. The fitting pass is run with gradients disabled and the model in evaluation mode. For each audio clip (up to $N_{\max}$ clips), a forward pass is executed with data collection enabled. The hook registered at block (ℓ, b) intercepts, for every head h, the tensor $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N^\ell_w \times M \times d_h}$ and flattens it along the spatial axes to obtain a matrix of shape $\mathbb{R}^{(N^\ell_w \cdot M) \times d_h}$, which is appended to an accumulation buffer for the triplet (ℓ, b, h). After all clips have been processed, the per-clip contributions are stacked to form the *fitting matrix*

$$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\text{fit}} \times d_h}, \qquad N_{\text{fit}} = N_{\text{clips}} \cdot N^\ell_w \cdot M, \quad (27)$$

where $N_{\text{clips}} \leq N_{\max}$ is the number of processed clips.

PCA decomposition. Standard PCA is applied to $\widetilde{\mathbf{R}}_{\ell,b,h}$. The empirical mean

$$\boldsymbol{\mu}_{\ell,b,h} = \frac{1}{N_{\text{fit}}} \sum_{i=1}^{N_{\text{fit}}} \widetilde{\mathbf{R}}_{\ell,b,h}[i,:] \in \mathbb{R}^{d_h} \quad (28)$$

is computed and subtracted, and the centred empirical covariance is decomposed as

$$\mathbf{C}_{\ell,b,h} = \frac{1}{N_{\text{fit}} - 1} \left(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^\top_{\ell,b,h} \right)^\top \left(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^\top_{\ell,b,h} \right)$$
$$= \boldsymbol{\Phi}_{\ell,b,h} \, \text{diag}(\lambda^{\text{ev}}_1, \ldots, \lambda^{\text{ev}}_{d_h}) \, \boldsymbol{\Phi}^\top_{\ell,b,h}, \quad (29)$$
$$\text{with } \lambda^{\text{ev}}_1 \geq \cdots \geq \lambda^{\text{ev}}_{d_h} \geq 0, \quad (30)$$

where $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ is the orthogonal matrix of eigenvectors (columns ordered by decreasing eigenvalue) and λ^{ev}_j denotes the j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$. We use the superscript "ev" throughout to distinguish these fixed quantities from the trainable weight vector $\boldsymbol{\lambda}_{\ell,b,h}$. Both $\boldsymbol{\Phi}_{\ell,b,h}$ and $\boldsymbol{\mu}_{\ell,b,h}$ are stored as non-trainable buffers; the accumulation buffers $\widetilde{\mathbf{R}}_{\ell,b,h}$ are then cleared to free memory.

Task-relevant subspace and initialisation of $\boldsymbol{\lambda}_{\ell,b,h}$. For each head (ℓ, b, h), the index $k_{\ell,b,h}$ defined in Eq. (**??**) identifies the smallest number of principal components that together explain at least a fraction ρ_{var} of the total variance,

providing a per-head estimate of the effective dimensionality of the task-relevant subspace. The learnable weight vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is initialised component-wise as

$$\lambda_j^{(0)} \;=\; \begin{cases} 1 & j \le k_{\ell,b,h}, \\ 0 & j > k_{\ell,b,h}, \end{cases} \tag{31}$$

corresponding to the identity map restricted to the task-relevant subspace and complete suppression of the remaining noise components. At this initialisation point, RD*therefore acts as a projection-and-reconstruction operator that retains the top-$k_{\ell,b,h}$ PCA directions unchanged and zeros out the remainder.

C.1.2. Optimisation of $\boldsymbol{\lambda}$ via Zero-Shot Accuracy

After the fitting pass has fixed the PCA bases $\{\boldsymbol{\Phi}_{\ell,b,h}\}$ and initialised $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ via Eq. (31), the second phase optimises the weight vectors to maximise zero-shot classification accuracy on a held-out validation set.

Parameter isolation. All model parameters except the weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}_{\ell\in\mathcal{L},\,b,\,h}$ are frozen by setting `requires_grad = False`. The only degrees of freedom during optimisation are therefore the $\sum_{\ell\in\mathcal{L}} B_\ell \cdot H_\ell$ weight vectors, each of dimension $d_h = 24$, that control the per-component gain of each attention head in the target stages $\mathcal{L}$. The CLAP audio projection head $P : \mathbb{R}^{D_3} \to \mathbb{R}^{d_{\text{proj}}}$ is kept frozen but is included in the computational graph so that gradients flow from the similarity space back to $\boldsymbol{\lambda}_{\ell,b,h}$.

Objective. Given a set of audio clips with ground-truth class labels $y \in \{1,\dots,C\}$ and pre-computed ℓ_2-normalised text embeddings $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$ for each class c, the optimisation objective is the cross-entropy loss over zero-shot logits:

$$\mathcal{J}(\boldsymbol{\lambda}) \;=\; -\,\mathbb{E}_{(x,\,y)}\left[\log \frac{\exp\!\big(\hat{\mathbf{a}}(x)^\top \mathbf{t}_y\big)}{\displaystyle\sum_{c=1}^{C} \exp\!\big(\hat{\mathbf{a}}(x)^\top \mathbf{t}_c\big)}\right], \tag{32}$$

where $\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$ is the ℓ_2-normalised audio embedding produced by P applied to the spatially averaged Stage-3 output of HTS-AT after the RD*transforms have been applied (cf. Table 3). Minimising $\mathcal{J}$ is equivalent to maximising zero-shot classification accuracy in expectation.

Optimiser and schedule. The weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are updated with AdamW (**?**) at learning rate η and weight decay γ; all hyperparameter values are reported in [PLACEHOLDER: §X / Table X]. The CLAP audio projection head $P : \mathbb{R}^{D_3} \to \mathbb{R}^{d_{\text{proj}}}$ is kept frozen but is included in the computational graph so that gradients flow from the similarity space back to $\boldsymbol{\lambda}_{\ell,b,h}$.

Early stopping and best-$\boldsymbol{\lambda}$ restoration. At the end of each epoch, zero-shot accuracy on the validation set is evaluated. A snapshot of the current $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ is saved whenever a new accuracy maximum is achieved. Training halts early if no improvement is observed for p consecutive epochs; at termination, the best snapshot is restored, guaranteeing that the final weight vectors correspond to the configuration achieving the highest observed validation accuracy.

Hyperparameter summary. Table 5 collects all hyperparameters introduced by RD*together with the default values used in all experiments.

Remark on hook-based extraction. The `AttentionHook` mechanism described in §B.1 for the dimensionality analysis is the same mechanism through which RD*applies reweighting at inference time. Hooks are registered once at model construction via `register_forward_hook` and remain permanently attached to the target `WindowAttention` modules throughout the entire model lifetime. The hook operates in two distinct modes controlled by an internal `collect_for_fitting` flag: in *collection mode* it accumulates raw head outputs $\mathbf{H}_{\ell,b,h}$ for PCA estimation without modifying the forward-pass output; in *reweighting mode* it replaces $\mathbf{H}_{\ell,b,h}$ with $\mathbf{H}^*_{\ell,b,h}$ (Eq. (36)) and re-applies the output projection $W^O_{\ell,b}$ and `proj_drop` before returning, since the hook return value supersedes the original `WindowAttention` output entirely.

C.1.3. Forward Pass with Spectral Reweighting

At inference time, the hook at block (ℓ, b) intercepts the raw head outputs $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ (cf. Eq. (6)) and applies the following sequence of operations independently for each head h.

Step 1 — Centering.

$$\widetilde{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h} \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{33}$$

where $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is broadcast over the first two axes.

Step 2 — Projection onto the full PCA basis.

$$\mathbf{Z}_{\ell,b,h} \;=\; \widetilde{\mathbf{H}}_{\ell,b,h}\,\boldsymbol{\Phi}_{\ell,b,h} \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}. \tag{34}$$

Unlike fixed-weight variants that truncate to $k_{\ell,b,h} < d_h$ components, RD*projects onto the *full* basis $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$, delegating the effective truncation to the learned weights.

Step 3 — Per-component rescaling.

$$\widehat{\mathbf{Z}}_{\ell,b,h} \;=\; \mathbf{Z}_{\ell,b,h} \odot \boldsymbol{\lambda}_{\ell,b,h}^{\top} \;\in\; \mathbb{R}^{N_w^{\ell}\times M\times d_h}, \tag{35}$$

where $\boldsymbol{\lambda}_{\ell,b,h}^{\top} \in \mathbb{R}^{d_h}$ is broadcast over the first two axes. Because $\boldsymbol{\lambda}_{\ell,b,h}$ is a free real-valued parameter, each entry λ_j can amplify ($|\lambda_j| > 1$), attenuate ($|\lambda_j| \in (0,1)$), suppress ($\lambda_j = 0$), or invert ($\lambda_j < 0$) the corresponding principal component.

Step 4 — Reconstruction in the original space.

$$\mathbf{H}^{*}_{\ell,b,h} \;=\; \widehat{\mathbf{Z}}_{\ell,b,h}\,\boldsymbol{\Phi}_{\ell,b,h}^{\top} \;\in\; \mathbb{R}^{N_w^{\ell}\times M\times d_h}. \tag{36}$$

The empirical mean $\boldsymbol{\mu}_{\ell,b,h}$ is *not* restored after reconstruction. Since $\mathbf{H}^{*}_{\ell,b,h}$ is immediately passed to the output projection $W^{O}_{\ell,b}$ (cf. Eq. (12)), which carries its own learnable bias $\mathbf{b}^{O}_{\ell,b} \in \mathbb{R}^{D_\ell}$, re-adding $\boldsymbol{\mu}_{\ell,b,h}$ would introduce a systematic, data-dependent shift in the residual stream that is already absorbed by $\mathbf{b}^{O}_{\ell,b}$.

Combining Steps 1–4, the full RD*transform reads:

$$\mathrm{RD}^{*}(\mathbf{H}_{\ell,b,h},\,\boldsymbol{\lambda}_{\ell,b,h}) \;=\; \big(\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}\big)\,\boldsymbol{\Phi}_{\ell,b,h}\;\mathrm{diag}(\boldsymbol{\lambda}_{\ell,b,h})\,\boldsymbol{\Phi}_{\ell,b,h}^{\top} \;\in\; \mathbb{R}^{N_w^{\ell}\times M\times d_h} \tag{37}$$

The tensor $\mathbf{H}^{*}_{\ell,b,h}$ replaces $\mathbf{H}_{\ell,b,h}$ in Eq. (12), so the output projection $W^{O}_{\ell,b}$ and the residual-stream update of Eq. (8) proceed unchanged. During optimisation, gradients of any downstream loss flow through RD* and update only $\boldsymbol{\lambda}_{\ell,b,h}$; the basis $\boldsymbol{\Phi}_{\ell,b,h}$, the mean $\boldsymbol{\mu}_{\ell,b,h}$, and all original model parameters are held fixed.

D. K-Fold Evaluation Pipeline and Results

D.1. Experimental Protocol

To obtain reliable and unbiased estimates of zero-shot classification accuracy, we adopt a stratified K-fold cross-validation protocol ($K = 5$) applied uniformly to both the CLAP baseline and ResiDual*. All experiments are conducted on four publicly available audio classification benchmarks: ESC-50 (**?**), IRMAS (**?**), TinySOL (**?**), and VocalSound (**?**), which collectively span environmental sounds, musical instruments, orchestral timbres, and human vocal categories.

Dataset sampling. For each dataset, up to $N_{\text{samples}} = 2{,}000$ examples are selected via class-stratified sampling with a fixed random seed, ensuring a balanced class distribution across all folds. The stratified sampling procedure first computes the per-class allocation $\lfloor N_{\text{samples}}/C \rfloor$ (plus a unit remainder distributed to the first $N_{\text{samples}} \bmod C$ classes), then samples without replacement from each class independently, guaranteeing reproducibility via a global seed.

Cross-validation split. The sampled subset is partitioned into $K = 5$ stratified folds using `StratifiedKFold` with `shuffle=True` and a fixed `random_state`. At each fold iteration, the training split (four folds) is further divided into two non-overlapping sub-partitions: a *PCA fitting* partition of $N_{\text{PCA}} = 250$ samples, used exclusively to estimate the spectral bases $\{\boldsymbol{\Phi}_{\ell,b,h},\,\boldsymbol{\mu}_{\ell,b,h}\}$ (cf. §C.1.1), and a λ-*optimisation* partition comprising the remaining training examples, used to refine the weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ via the objective of Eq. (32). The held-out fold (one fifth of the sampled subset) is reserved exclusively for final zero-shot evaluation.

Text prompt construction. Class-conditional text embeddings are computed once per dataset prior to the fold loop. Each class label c is formatted as the natural-language prompt `"this is the sound of {c}"`, where underscores are replaced by spaces. This prompt template follows the zero-shot inference convention adopted in the official Microsoft CLAP repository (**?**), where the same phrasing is used for audio classification evaluation across multiple benchmarks. The resulting prompts are encoded by the frozen CLAP text encoder and ℓ_2-normalised to obtain $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$ for each class $c \in \{1,\ldots,C\}$.

Zero-shot inference. At evaluation time, the audio embedding $\hat{\mathbf{a}}(x)$ for a test clip x is obtained by passing the raw waveform through the (reweighted) audio encoder followed by the projection head P, and ℓ_2-normalising the result. The predicted class is then

$$\hat{y}(x) \;=\; \arg\max_{c\in\{1,\ldots,C\}} \hat{\mathbf{a}}(x)^{\top}\mathbf{t}_c. \tag{38}$$

For the CLAP baseline, no adaptation is performed and the frozen model is evaluated directly. For ResiDual*, the per-fold PCA fitting and λ-optimisation phases precede evaluation, and the model weights $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are fully reset between folds by reloading the CLAP checkpoint and re-instantiating all `SpectralReweightingLayer` modules.

Optimiser configuration. The weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are updated with AdamW (**?**) with learning rate $\eta = 5\times10^{-3}$ and weight decay $\gamma = 5\times10^{-2}$, for a maximum of $T_{\max} = 40$ epochs with early stopping patience $p = 5$. The target stages are $\mathcal{L} = \{2,3\}$ and the PCA variance threshold is $\rho_{\text{var}} = 0.95$. All other hyperparameters follow Table 5.

D.2. Quantitative Results

Table 6 reports the mean zero-shot accuracy and standard deviation across the five folds for each dataset and each model variant. Bold entries indicate the higher mean accuracy for each dataset. ResiDual*improves over the CLAP

baseline on all four benchmarks, with gains ranging from $+2.4$ percentage points on ESC-50 to $+54.8$ percentage points on TinySOL.

Table 7 provides the per-fold breakdown, offering a more granular view of stability across splits.

D.3. Visualisation and Discussion

Figures 6–9 provide complementary perspectives on the evaluation results.

Aggregate accuracy (Figure 6). Figure 6 summarises the mean zero-shot accuracy and fold-level variability (shown as ± 1 standard deviation error bars) for both model variants across the four benchmarks. The improvements conferred by ResiDual*are consistent and statistically robust, with the gain being most pronounced on TinySOL ($+54.8$ pp) and IRMAS ($+21.2$ pp) — the two datasets where the CLAP baseline suffers most, likely due to a structural mismatch between the fine-grained timbral distinctions required by these tasks and the generic audio-language alignment captured by the frozen CLAP model. On the semantically richer ESC-50 and VocalSound benchmarks, where CLAP already achieves solid baseline performance, ResiDual*continues to improve results, suggesting that spectral reweighting is additive even in near-saturating regimes.

Per-fold stability (Figure 7). Figure 7 traces the per-fold accuracy trajectory for each dataset. For both model variants the fold-to-fold variance is low (standard deviation ≤ 1.77 pp in all cases), confirming that neither the baseline nor the adapted model exhibits systematic sensitivity to the particular fold partition. The consistently higher ResiDual*trajectory across all folds and all datasets rules out the possibility that the reported gains arise from favourable fold splits.

Multi-benchmark overview (Figure 8). The radar chart in Figure 8 places all four datasets on the same polar axes, making the relative profile of each model immediately apparent. The CLAP Classic polygon is considerably smaller on the IRMAS and TinySOL axes, reflecting the difficulty of zero-shot transfer to fine-grained timbral classification tasks. The ResiDual*polygon is substantially larger on every axis, and notably more balanced, approaching the ESC-50 accuracy level on TinySOL — a benchmark on which the baseline achieves barely above chance (35.6% vs. 90.3%). This suggests that the spectral reweighting mechanism is especially effective at realigning internal representations toward task-relevant subspaces that the general-purpose CLAP model underutilises for timbral discrimination.

Lambda optimisation dynamics (Figure 9). Figure 9 shows the per-epoch validation accuracy on the λ-optimisation partition, averaged over the five folds (solid line), with individual fold trajectories overlaid in transparency and a ± 1 standard deviation band. The dashed vertical line and annotated dot mark the average best epoch across folds.

Several consistent patterns emerge across datasets. First, the optimisation trajectories exhibit a broadly monotone increasing trend in the early epochs, though with non-negligible plateau phases and occasional minor regressions in later epochs — a behaviour consistent with AdamW on a low-dimensional, non-convex loss landscape. Second, the average best epoch varies substantially across benchmarks: ESC-50 converges earliest (mean best epoch ≈ 21, though with high inter-fold variance ranging from epoch 11 to epoch 32), while IRMAS, TinySOL, and VocalSound all require a larger number of gradient steps (mean best epochs $\approx$ 37, 38, and 33 respectively). This pattern correlates with the absolute accuracy gain: benchmarks where ResiDual*achieves the largest improvement over the CLAP baseline (TinySOL $+54.8$ pp, IRMAS $+21.2$ pp) are precisely those where the optimiser must explore the λ-space more extensively before identifying a high-performing configuration. Third, the inter-fold variance of the optimisation trajectories is lowest for ESC-50 in terms of final accuracy, but highest in terms of convergence epoch (fold best epochs span 11–32), suggesting that the optimisation landscape for ESC-50 has multiple near-equivalent basins reachable at different depths. IRMAS and TinySOL exhibit the opposite profile: the convergence epoch is consistently late across folds (most folds reach or approach the maximum of 40 epochs), indicating a flatter landscape where marginal improvements accumulate slowly but reliably throughout training.

The gap between the best validation accuracy on the λ-optimisation partition and the held-out test accuracy (Table 6) varies considerably across benchmarks and folds. On ESC-50, the gap is consistently small (≤ 2.5 pp across all folds), indicating near-perfect generalisation of the optimised λ configuration to unseen audio within the same distribution. On IRMAS and VocalSound, however, the gap is substantially larger, reaching up to ≈ 11 pp on individual IRMAS folds and ≈ 10 pp on VocalSound fold 3. This discrepancy between optimisation-partition accuracy and held-out test accuracy does not necessarily imply overfitting of the λ weights in the classical sense: the λ-optimisation partition and the test set are drawn from the same dataset but constitute non-overlapping stratified splits, and the gap likely reflects the limited size of each partition ($\approx$1,350 samples) combined with the inherent difficulty of zero-shot transfer for fine-grained timbral tasks such as IRMAS and VocalSound. Importantly, despite this

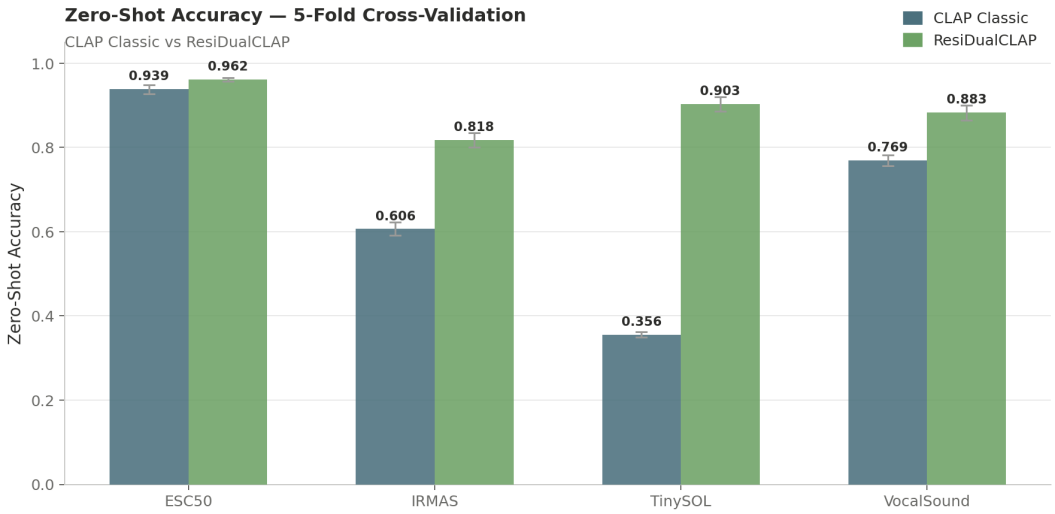


Figure 6. **Mean zero-shot accuracy across benchmarks.** Bar heights correspond to the mean accuracy over the 5 folds; error bars denote ± 1 standard deviation. CLAP Classic (steel blue) vs. ResiDual*(green).

gap, the held-out test accuracy of ResiDual*remains substantially above the CLAP baseline on all benchmarks and all folds, confirming that the spectral reweighting generalises beyond the optimisation set even when the gap is non-negligible.

D.4. Summary

The optimisation dynamics are consistent with the viability of the RD*pipeline: convergence is broadly monotone in the early training phase, and early stopping reliably identifies configurations that substantially outperform the frozen baseline on held-out data. The gap between optimisation-partition and test accuracy is small on ESC-50 but more pronounced on IRMAS and VocalSound, pointing to a generalisation challenge that scales with task difficulty rather than to overfitting of the λ parameters per se. These results collectively validate the design choices described in Appendix C: a fixed PCA basis estimated on a small unlabelled subset, combined with a lightweight per-component rescaling optimised via cross-entropy on a modest labelled partition, suffices to substantially improve zero-shot classification accuracy without modifying any pretrained parameters, even under conditions of imperfect optimisation-to-test generalisation.

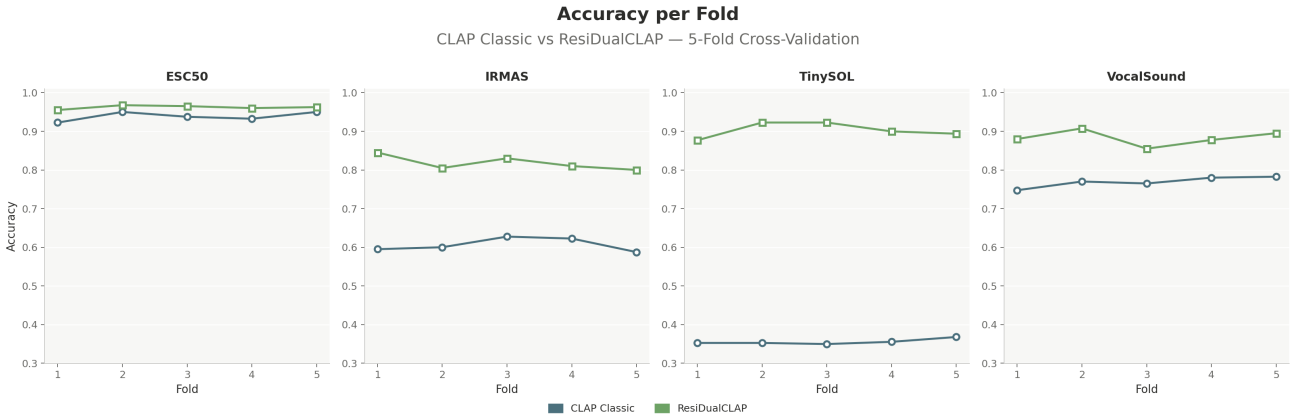


Figure 7. **Fold-by-fold accuracy trajectories.** Each panel corresponds to one benchmark. Circles (CLAP Classic) and squares (ResiDual*) mark individual fold accuracies; lines connect consecutive folds to aid trend visualisation. The y-axis range is shared across folds within each panel but scaled independently per dataset to emphasise within-dataset variance.

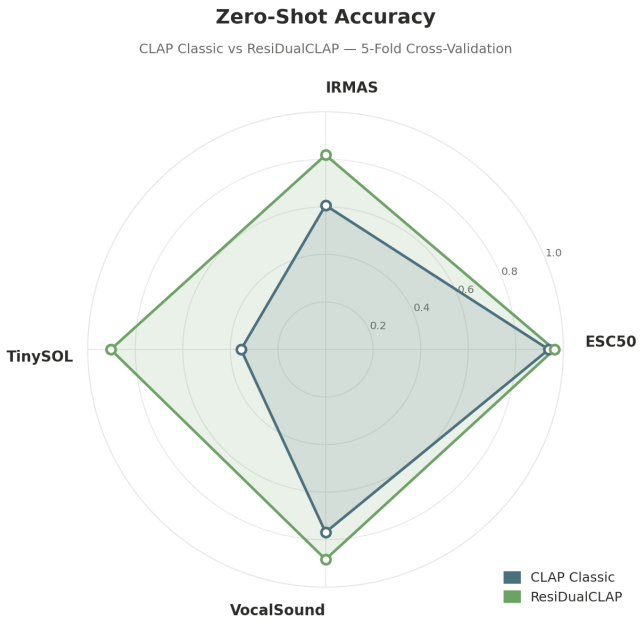


Figure 8. **Radar chart of zero-shot accuracy across benchmarks.** Each axis corresponds to one of the four evaluation datasets. Mean 5-fold accuracy is used as the radial coordinate. The shaded regions indicate the area enclosed by each model's accuracy profile.

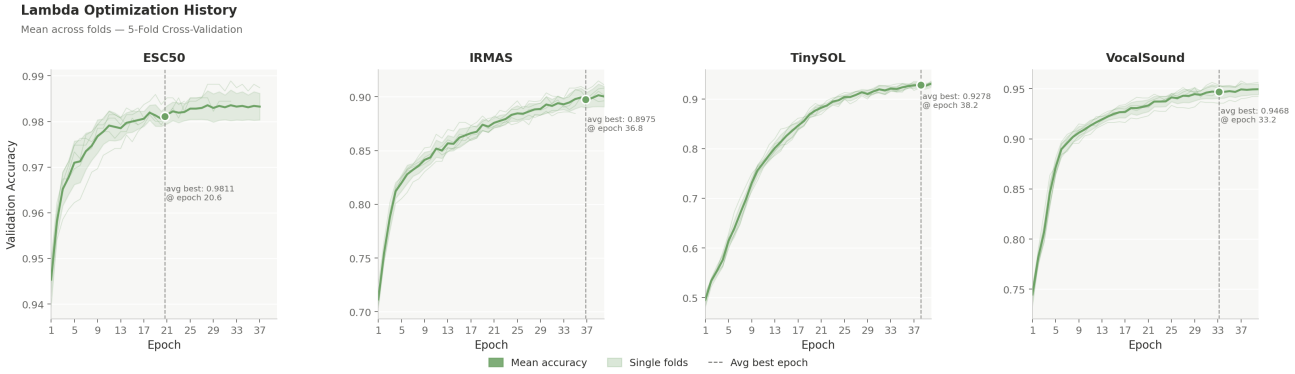


Figure 9. **Lambda optimisation history.** Each panel shows the validation accuracy on the λ-optimisation partition as a function of epoch, for one benchmark. The solid line is the fold-mean; the shaded band is ± 1 standard deviation; semi-transparent lines show individual fold trajectories. The dashed vertical line and annotated marker indicate the average best epoch (where the snapshot used for evaluation was taken).

Table 3. Summary of notation used throughout the paper. Symbols introduced by RD* are marked with $(\star)$.

Symbol	**Definition**
$\ell \in \{0, 1, 2, 3\}$	Stage index
$b \in \{1, \ldots, B_\ell\}$	Block index within stage ℓ
$h \in \{1, \ldots, H_\ell\}$	Attention head index within block (ℓ, b)
S_ℓ	Spatial side length of the token grid at stage ℓ
$w = 8$	Attention window side length
$M = w^2 = 64$	Tokens per attention window
$N_w^\ell = S_\ell^2 / M$	Number of attention windows at stage ℓ
$d_h = 24$	Per-head feature dimension (constant across stages)
$D_\ell = H_\ell \cdot d_h$	Total embedding dimension at stage ℓ
$d_{\text{proj}} = 1024$	CLAP joint embedding dimension
$W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$	Fused QKV projection at block (ℓ, b)
$W_{\ell,b}^{O} \in \mathbb{R}^{D_\ell \times D_\ell}$	Output projection at block (ℓ, b)
$W_{\ell,b,h}^{O} \in \mathbb{R}^{d_h \times D_\ell}$	Row slice of $W_{\ell,b}^{O}$ for head h
$\mathbf{b}_{\ell,b}^{O} \in \mathbb{R}^{D_\ell}$	Output projection bias at block (ℓ, b)
$\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$	Relative position bias for head h at block (ℓ, b)
$\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Raw head output at block (ℓ, b), head h (pre-$W_{\ell,b}^{O}$)
$\widehat{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$	Per-head projected contribution (post-$W_{\ell,b}^{O}$); see Eq. (14)
$\mathbf{r}_{\ell,b,h} \in \mathbb{R}^{d_h}$	Spatially aggregated raw head output; see Eq. (15)
$\widehat{\mathbf{r}}_{\ell,b,h} \in \mathbb{R}^{D_\ell}$	Spatially aggregated projected head output; see Eq. (**??**)
$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$	Dataset matrix of $\mathbf{r}_{\ell,b,h}$ across n samples
$\widehat{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n \times D_\ell}$	Dataset matrix of $\widehat{\mathbf{r}}_{\ell,b,h}$ across n samples
$P : \mathbb{R}^{768} \to \mathbb{R}^{1024}$	CLAP audio projection head (two-layer MLP with residual)
n	Number of audio samples in the dataset
Symbols introduced by RD* $(\star)$	
$\mathcal{L} \subseteq \{0, 1, 2, 3\}$ $(\star)$	Target stage indices for reweighting
$\rho_{\text{var}} \in (0, 1)$ $(\star)$	Variance threshold for per-head subspace estimation (Eq. (**??**))
$k_{\ell,b,h} \in \{1, \ldots, d_h\}$ $(\star)$	Number of task-relevant PCA components for head h at block (ℓ, b), determined by ρ_{var} (Eq. (**??**))
$\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ $(\star)$	Learnable per-component weight vector for head h at block (ℓ, b); initialised via Eq. (31), optimised via Eq. (32
$\mathbf{C}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$	Empirical covariance matrix of $\widetilde{\mathbf{R}}_{\ell,b,h}$
$\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$	Orthogonal matrix of eigenvectors of $\mathbf{C}_{\ell,b,h}$
λ_j^{ev}	j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$, $\lambda_1^{\text{ev}} \geq \cdots \geq \lambda_{d_h}^{\text{ev}} \geq 0$

Table 4. **Selezione delle teste più specializzate (Z-score coerenza).** Per brevità è riportata solo la prima descrizione selezionata dall'OMP; i risultati completi e i mapping tra indici assoluti e gerarchici sono consultabili nel notebook di analisi su GitHub.

Head ID	Layer	Dominant Cat.	Coherence Z	PC1 Var.	Primary Desc.
L3_B0_H7	3	*human_body*	1.57	0.13	baby crying
L2_B4_H1	2	*human_body*	1.46	0.11	person sneezing
L2_B5_H15	2	*nature*	1.23	0.14	water dripping
L3_B0_H25	3	*nature*	1.19	0.19	water dripping
L3_B1_H9	3	*urban*	0.79	0.16	chainsaw cutting
L2_B5_H6	2	*animals*	0.71	0.14	frog croaking
L3_B1_H17	3	*animals*	0.65	0.29	rooster crowing

Table 5. RD*hyperparameters and default values.

Symbol	Description	Default
$\mathcal{L}$	Target stage indices	$\{1, 2, 3\}$
ρ_{var}	Variance threshold for $k_{\ell,b,h}$ (Eq. (**??**))	0.95
η	Learning rate (AdamW)	[PLACEHOLDER: §X]
γ	Weight decay (AdamW)	[PLACEHOLDER: §X]
p	Early-stopping patience (epochs)	5
$T_{\max}$	Maximum optimisation epochs	30
$N_{\max}$	Maximum fitting clips	$10\,000$

Table 6. Zero-shot accuracy (%) under 5-fold cross-validation. Mean $\pm$ standard deviation across folds. Bold indicates the higher accuracy for each dataset.

Dataset	Classes	Samples	CLAP Classic	ResiDual*
ESC-50	50	2 000	93.85 ± 1.06	$\mathbf{96.20 \pm 0.43}$
IRMAS	11	2 000	60.65 ± 1.57	$\mathbf{81.80 \pm 1.69}$
TinySOL	14	1 913†	35.55 ± 0.64	$\mathbf{90.31 \pm 1.76}$
VocalSound	6	2 000	76.90 ± 1.25	$\mathbf{88.30 \pm 1.77}$

†TinySOL contains 1 913 valid samples after validation; all are used.

Table 7. Per-fold zero-shot accuracy (%) for CLAP Classic and ResiDual*. Bold entries indicate the higher accuracy within each fold and dataset.

Dataset	Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
ESC-50	CLAP Classic	92.25	95.00	93.75	93.25	95.00
	ResiDual*	**95.50**	**96.75**	**96.50**	**96.00**	**96.25**
IRMAS	CLAP Classic	59.50	60.00	62.75	62.25	58.75
	ResiDual*	**84.50**	**80.50**	**83.00**	**81.00**	**80.00**
TinySOL	CLAP Classic	35.24	35.24	34.96	35.53	36.78
	ResiDual*	**87.68**	**92.26**	**92.26**	**89.97**	**89.37**
VocalSound	CLAP Classic	74.75	77.00	76.50	78.00	78.25
	ResiDual*	**88.00**	**90.75**	**85.50**	**87.75**	**89.50**